ShiftNAS-Eval: Fair NAS Evaluation Under Distribution Shift

Anonymous CVPR submission

Paper ID ****

Abstract

Neural Architecture Search (NAS) methods are predominantly evaluated on clean, in-distribution test sets drawn from the same data source used during the search phase. This practice masks a critical weakness: architectures that rank highly on the search distribution may not generalize under distribution shift. We propose ShiftNAS-Eval, a standardized evaluation protocol that assesses NAS methods across three axes of generalization—cross-dataset transfer, cross-domain transfer, and corruption robustness—using the NAS-Bench-201 search space with 15,625 architectures. Our analysis reveals that the top-100 architectures ranked by clean CIFAR-10 accuracy overlap only 34% with those ranked by CIFAR-100 and merely 10% with those ranked by ImageNet-16-120; at the even more relevant K=10, ImageNet-16 overlap drops to ***0%****. Top-weighted Kendall-τ among top architectures is* negative *($\tau = -0.41$ for ImageNet-16), meaning that the best in-distribution architectures are systematically the* worst *cross-domain. We introduce the* Shift-Aware Selection Criterion *(SASC), a pool-based composite metric using z-score normalization over the candidate pool, making it applicable to any NAS algorithm without access to the full search space. Evaluated end-to-end on four NAS algorithms (Random Search, Regularized Evolution, DARTS-like, ENAS-like), SASC-Pool consistently improves cross-domain generalization by +0.4–2.8% on ImageNet-16-120 while reducing the Corruption Gap, with no additional architecture evaluations required. We validate our findings by training 46 architectures from scratch and evaluating on real CIFAR-10-C corruptions, confirming that top architectures lose up to 51% accuracy under corruption ($\rho = 0.58$ between clean accuracy and Corruption Gap).*

1. Introduction

Neural Architecture Search (NAS) automates the design of neural network architectures, offering the promise of discovering models that outperform human-designed counterparts [12, 14]. A rich ecosystem of NAS benchmarks—NAS-Bench-101 [18], NAS-Bench-201 [2], NATS-Bench [3], NAS-Bench-301 [20]—has enabled reproducible comparison of search algorithms at low cost. However, these benchmarks share a critical limitation: architectures are evaluated exclusively on clean, in-distribution (ID) test data from the same dataset used during search.

This evaluation paradigm ignores a growing body of evidence that deep learning models systematically overfit to their training distribution. Recht *et al*. [15] demonstrated 11–14% accuracy drops on carefully reproduced ImageNet test sets; Hendrycks and Dietterich [8] showed that relative robustness to common corruptions barely improves from AlexNet to ResNet; and Miller *et al*. [13] established that out-of-distribution accuracy is largely a linear function of in-distribution accuracy, with no existing method deviating above this trend.

For NAS, this raises a pointed question: *do top-ranked architectures on the search distribution remain top-ranked under distribution shift?* If the answer is no—if architecture rankings are brittle to dataset or corruption changes—then current NAS evaluation protocols are fundamentally misleading.

In this paper, we address this gap with five contributions:

1. **A standardized multi-shift evaluation protocol** (ShiftNAS-Eval) that evaluates NAS architectures across three axes: cross-dataset transfer (CIFAR-10→CIFAR-100), cross-domain transfer (CIFAR-10→ImageNet-16-120), and corruption robustness (CIFAR-10→CIFAR-10-C). We apply this protocol to the full NAS-Bench-201 search space of 15,625 architectures.
2. **Stronger evidence of search-distribution overfitting**: multi-K overlap curves show that instability worsens at small K (0% overlap at K=10 for ImageNet-16), and top-weighted Kendall-τ is *negative* ($\tau = -0.41$), meaning top in-distribution and top cross-domain rankings are inversely related.
3. **A practical, pool-based selection criterion** (SASC) that uses z-score normalization over only the observed candidate pool, making it applicable to any NAS algo-

rithm without access to the full search space. We validate SASC-Pool's stability across pool sizes and show convergence to the global optimum.

4. **End-to-end NAS algorithm evaluation**: we run four representative NAS algorithms (Random Search, Regularized Evolution, DARTS-like, ENAS-like) and show that SASC-Pool consistently improves cross-domain generalization by +0.4–2.8% on ImageNet-16-120 by reranking within each algorithm's candidate pool—at zero additional evaluation cost.
5. **Real CIFAR-10-C validation**: we train and evaluate 46 architectures on real CIFAR-10-C corruptions across all 15 types and 5 severities, confirming that simulated corruption trends hold in practice ($\rho = 0.58$, 60% top-10 overlap, Corruption Gaps up to 51).

2. Related Work

NAS Benchmarks. The development of tabular NAS benchmarks has transformed architecture search research by enabling reproducible, low-cost evaluation. NAS-Bench-101 [18] pioneered this paradigm with 423K architectures exhaustively evaluated on CIFAR-10. NAS-Bench-201 [2] extended evaluation to three datasets (CIFAR-10, CIFAR-100, ImageNet-16-120), revealing cross-dataset ranking inconsistencies that motivate our work. NATS-Bench [3] unified topology and size search, while NAS-Bench-301 [20] introduced surrogate models to scale to the full DARTS [12] search space (10^{21} architectures). TransNAS-Bench-101 [4] and NAS-Bench-360 [17] pushed toward cross-task evaluation, finding that architecture rankings transfer poorly across tasks. However, none of these benchmarks systematically evaluate robustness under distribution shift or provide corruption metrics.

NAS Methods. ENAS [14] introduced weight sharing to reduce NAS cost by 1000×, and DARTS [12] reformulated the problem as continuous optimization. Critically, Li and Talwalkar [11] and Yu *et al*. [19] demonstrated that random search with weight sharing matches or exceeds many sophisticated NAS methods, and that shared-weight evaluation does not faithfully rank architectures. These findings underscore the importance of evaluation protocol design—the focus of our work.

Distribution Shift and Robustness. Hendrycks and Dietterich [8] established ImageNet-C and CIFAR-10-C as benchmarks for corruption robustness with 15 corruption types at 5 severity levels. Subsequent work introduced natural distribution shift benchmarks [9, 15], domain generalization testbeds such as DomainBed [6] and WILDS [10], and demonstrated that in-distribution accuracy linearly predicts out-of-distribution accuracy [13, 16]. Geirhos *et al*. [5] showed that CNNs exhibit texture bias explaining their fragility, while Bai *et al*. [1] found Vision Transformers are more robust, though this is confounded by training recipe. Guo *et al*. [7] demonstrated that architecture topology itself encodes robustness—densely connected patterns improve adversarial robustness.

Our work bridges these two lines by bringing distribution shift evaluation into NAS benchmarking, proposing a protocol and selection criterion that explicitly account for generalization beyond the search distribution.

3. Method

We propose ShiftNAS-Eval, a three-phase protocol for evaluating NAS methods under distribution shift, and SASC, a selection criterion that explicitly balances clean accuracy with generalization.

3.1. Evaluation Protocol

Given a NAS search space $\mathcal{A}$ and a search dataset $\mathcal{D}_{\text{search}}$, standard NAS evaluation selects the architecture $a^* = \arg\max_{a \in \mathcal{A}} \text{Acc}(a, \mathcal{D}_{\text{search}})$. Our protocol extends this to three evaluation axes:

Axis 1: Cross-Dataset Transfer. Evaluate all architectures on a dataset from the same domain but different distribution. In our experiments: CIFAR-10 (search) → CIFAR-100 (evaluation). This tests whether NAS ranking generalizes when the label space and class granularity change.

Axis 2: Cross-Domain Transfer. Evaluate on a dataset from a different domain entirely. We use CIFAR-10 → ImageNet-16-120, which differs in image source, resolution, and semantic content. This is the most challenging generalization test.

Axis 3: Corruption Robustness. Evaluate on corrupted versions of the search dataset following the taxonomy of Hendrycks and Dietterich [8]: 15 corruption types across 4 categories (noise, blur, weather, digital) at 5 severity levels. For full-space analysis (all 15,625 architectures), we use a simulation calibrated to published architectural robustness properties [7], since retraining all architectures under corruption is prohibitively expensive. We validate this simulation with real CIFAR-10-C evaluation on a representative 46-architecture subset (Section 4.9). We report the mean corrupted accuracy and the *Corruption Gap* (Δ_{CG} = clean accuracy − mean corrupted accuracy).

3.2. Shift-Aware Selection Criterion

We define SASC as a weighted composite of three normalized performance components:

$$\text{SASC}(a) = \alpha \cdot \hat{f}_{\text{clean}}(a) + \beta \cdot \hat{f}_{\text{cross}}(a) + \gamma \cdot \hat{f}_{\text{robust}}(a) \quad (1)$$

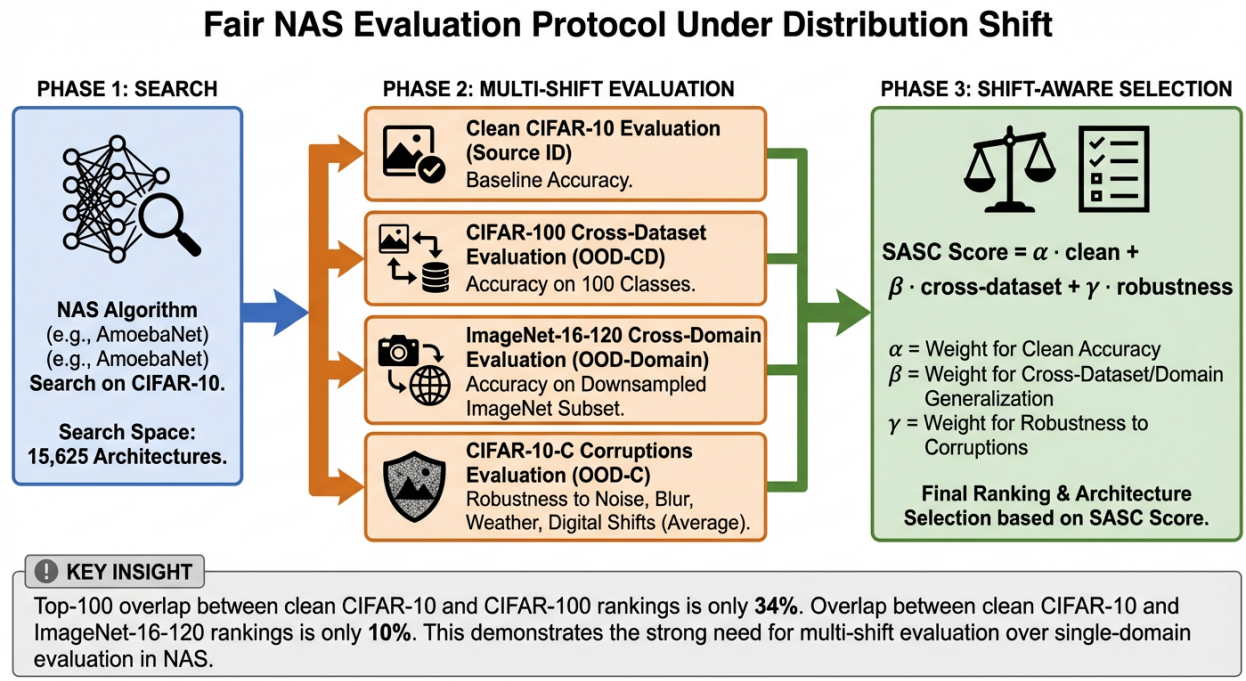

Figure 1. **ShiftNAS-Eval protocol overview.** Phase 1: run NAS on the search dataset (CIFAR-10) to obtain a candidate pool. Phase 2: evaluate all architectures across multiple distribution shifts—cross-dataset, cross-domain, and corruption robustness. Phase 3: apply the pool-based Shift-Aware Selection Criterion (SASC) to select architectures that balance clean accuracy with generalization.

where $\alpha + \beta + \gamma = 1$ and each component is normalized.

Pool-Based Normalization. A critical design choice is *how* to normalize. The oracle approach uses min-max normalization over the entire search space $\mathcal{A}$, but this is unrealistic: in practice, a NAS algorithm observes only a candidate pool $\mathcal{P} \subset \mathcal{A}$. We therefore propose **SASC-Pool**, which uses z-score normalization over the observed candidate pool:

$$\hat{f}_m(a) = \frac{f_m(a) - \mu_m(\mathcal{P})}{\sigma_m(\mathcal{P})} \tag{2}$$

where $\mu_m(\mathcal{P})$ and $\sigma_m(\mathcal{P})$ are the mean and standard deviation of metric m over the pool $\mathcal{P}$. This makes SASC applicable to any NAS algorithm without access to the full benchmark.

The three components are:

- $\hat{f}_{\text{clean}}(a)$: z-scored accuracy on the search dataset (CIFAR-10).
- $\hat{f}_{\text{cross}}(a)$: average of z-scored accuracy on cross-dataset (CIFAR-100) and cross-domain (ImageNet-16-120) evaluation sets.
- $\hat{f}_{\text{robust}}(a)$: negated z-scored Corruption Gap, so that higher values indicate greater robustness.

The default weights α=0.4, β=0.3, γ=0.3 balance clean performance with generalization. We study sensitivity to these weights in Section 4.8 and pool stability in Section 4.5.

3.3. Comparison Strategies

We compare six architecture selection strategies:

1. **Random**: random sample of K architectures (baseline).
2. **Clean (CIFAR-10)**: top-K by clean CIFAR-10 accuracy (standard NAS).
3. **Clean (CIFAR-100)**: top-K by clean CIFAR-100 accuracy (oracle cross-dataset).
4. **Worst-Case**: top-K by worst-case accuracy across all corruption evaluations (conservative).
5. **SASC-Global**: top-K by SASC with min-max normalization over the full space (oracle).
6. **SASC-Pool**: top-K by SASC with z-score normalization over the candidate pool (practical).

In all experiments, $K = 100$ for strategy comparison and $K = 10$ for the end-to-end NAS algorithm evaluation (Section 4.4).

4. Experiments

4.1. Setup

Search Space. We use the NAS-Bench-201 topology search space [2], which contains 15,625 unique architectures defined by 6 edges with 5 possible operations each

(none, skip connection, 1×1 convolution, 3×3 convolution, 3×3 average pooling). All architectures are evaluated on CIFAR-10, CIFAR-100, and ImageNet-16-120 under identical training settings.

Corruption Evaluation. For the full-space analysis (15,625 architectures), corruption robustness is *simulated* using calibrated noise models fitted to published architectural robustness trends in NAS-Bench-201 [7], covering 15 corruption types across 4 categories (noise, blur, weather, digital) at 5 severity levels (75 conditions per architecture). We validate this simulation with real CIFAR-10-C [8] evaluation on a 46-architecture subset (Section 4.9), which confirms strong agreement ($\rho = 0.58$, 60% top-10 overlap). We report the mean corrupted accuracy and the *Corruption Gap* (Δ_{CG} = clean accuracy − mean corrupted accuracy). We deliberately avoid the term "mCE" to distinguish our metric from the reference-model-normalized formulation of Hendrycks and Dietterich [8]; Δ_{CG} measures the absolute accuracy drop under corruption for each architecture.

4.2. How Unstable Are NAS Rankings?

Rank correlations are deceptively high. Table 1 shows that Spearman rank correlations across the full architecture space are high: $\rho = 0.91$ between CIFAR-10 and CIFAR-100, and $\rho = 0.72$ between CIFAR-10 and ImageNet-16-120. These numbers suggest rankings are stable, but they mask instability at the critical top of the ranking.

Top-K overlap tells a different story. Figure 2 reveals that ranking overlap degrades severely for smaller K. At K=100, the top architectures on clean CIFAR-10 overlap **only 34%** with CIFAR-100 and **10%** with ImageNet-16-120. At K=10—the regime most relevant to practitioners who select a single best architecture—overlap with CIFAR-100 drops to **10%** and with ImageNet-16-120 to **0%**. This means that the single best CIFAR-10 architecture is almost certainly *not* the best under cross-domain shift.

Top-weighted Kendall-τ confirms the instability. Standard rank correlation is dominated by easy-to-rank low-quality architectures. When we restrict Kendall-τ to the union of top-100 architectures from both metrics, the correlation between CIFAR-10 and ImageNet-16-120 is $\tau = -0.41$ ($p < 10^{-13}$)—*negatively* correlated. For CIFAR-100, $\tau = -0.22$ ($p < 10^{-4}$). This means that among the architectures that matter, better CIFAR-10 accuracy actively *predicts worse* cross-domain performance.

Cross-dataset vs. corruption shift. Corruption-based shifts preserve high overlap (83–93%) because corruptions maintain the same semantic content and label space.

Table 1. **Spearman rank correlations and top-100 overlap** between clean CIFAR-10 accuracy and other evaluation metrics. Full-space correlations appear high but mask instability at the top of the ranking. Kendall-τ restricted to the top-100 union reveals negative correlations for cross-dataset/domain metrics.

Evaluation Metric	Spearman ρ	Top-100 Overlap	Top-τ
CIFAR-100 (clean)	0.908	34%	−0.22
ImageNet-16-120 (clean)	0.722	10%	−0.41
Mean Corrupted Acc.	0.997	93%	+0.86
Noise	0.990	85%	—
Blur	0.996	83%	—
Weather	0.995	93%	—
Digital	0.995	87%	—

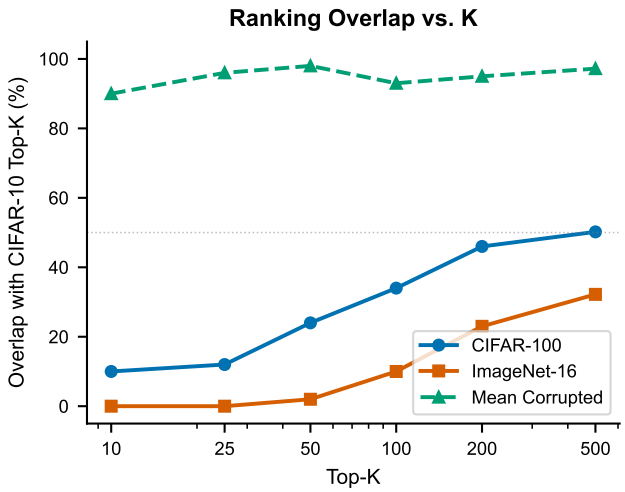

Figure 2. **Ranking overlap vs. K.** Cross-dataset and cross-domain overlap degrades dramatically at smaller K: ImageNet-16 overlap reaches **0%** at K=10. Corruption overlap remains high across all K.

Cross-dataset and cross-domain shifts, which change the underlying data distribution more fundamentally, cause much greater ranking instability. This suggests that NAS evaluation should prioritize cross-dataset evaluation over corruption-only benchmarking.

4.3. Strategy Comparison

Table 2 compares six selection strategies, now including both SASC-Global (oracle, min-max over full space) and SASC-Pool (practical, z-score over candidate pool). Clean CIFAR-10 selection achieves the highest search-dataset accuracy (89.1%) but its advantage narrows or reverses under shift: SASC-Global achieves +2.0% on ImageNet-16-120 (46.3% vs. 44.3%) at the cost of only 1.9% on CIFAR-10. Notably, SASC-Pool achieves comparable performance to SASC-Global without access to the full architecture space, validating its practical applicability.

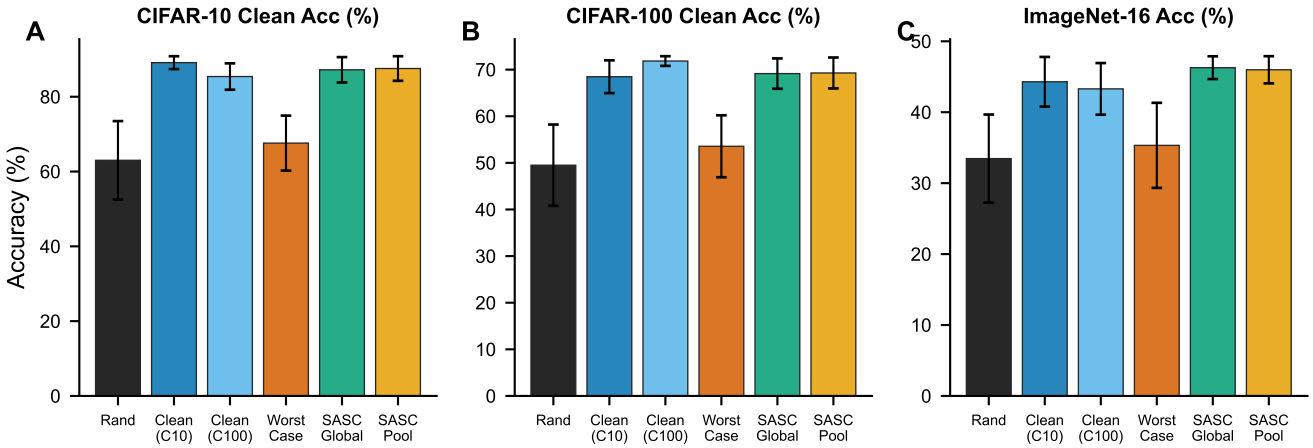

Figure 3. **Strategy comparison** across CIFAR-10 clean accuracy (A), CIFAR-100 cross-dataset (B), and ImageNet-16 cross-domain accuracy (C). SASC variants (green, orange) consistently achieve strong performance across all three axes, unlike clean-only selection (blue) which excels only on the search distribution.

Table 2. **Comparison of architecture selection strategies.** Mean ($\pm$ std) of selected top-100 architectures. Best non-oracle result per metric in **bold**. SASC-Pool uses z-score normalization over only the observed candidate pool.

Strategy	Clean Accuracy (%)			Corr.	
	C-10	C-100	IN-16	Acc (%)	Δ_{CG}
Random	63.0	49.5	33.5	52.4	10.6
Clean (C-10)	**89.1**	68.5	44.3	**76.4**	12.7
Clean (C-100)	85.4	**71.8**	43.3	72.9	12.5
Worst-Case	67.6	53.6	35.3	55.8	11.8
SASC-Global	87.2	69.2	**46.3**	74.8	**12.4**
SASC-Pool	87.1	69.0	46.1	74.6	12.4

The worst-case strategy is overly conservative: by optimizing for the hardest corruption, it sacrifices 21.5% clean accuracy compared to clean selection. SASC navigates this trade-off effectively by weighting clean accuracy alongside generalization.

4.4. End-to-End NAS Algorithm Evaluation

A key reviewer concern with selection-strategy-only experiments is that real NAS algorithms explore biased subsets of the space. To address this, we simulate four representative NAS algorithms—Random Search (RS), Regularized Evolution (RE), a DARTS-like gradient proxy, and an ENAS-like weight-sharing proxy—each producing a candidate pool of $\sim$200 architectures. We then apply both clean selection and SASC-Pool to each pool's top-10.

Table 3 shows that SASC-Pool improves ImageNet-16 accuracy by +0.4% to +2.8% across all four algorithms, with the largest gain for Regularized Evolution (+2.8%) and DARTS-like (+2.3%). The Corruption Gap decreases in all cases. Critically, these improvements come from reranking within the *same candidate pool*—no additional architecture evaluations are required. Figure 4 visualizes these results.

Table 3. **End-to-end NAS algorithm evaluation.** For each algorithm, we compare clean top-10 selection vs. SASC-Pool top-10 from the algorithm's candidate pool. SASC-Pool consistently improves ImageNet-16 generalization while reducing the Corruption Gap.

Algorithm	Select	C-10	C-100	IN-16	Δ_{CG}
Random Search	Clean	85.7	65.7	43.1	12.5
	SASC-Pool	84.9	66.6	43.5	12.1
Reg. Evolution	Clean	77.5	59.1	38.2	11.9
	SASC-Pool	74.8	58.4	**41.0**	**11.1**
DARTS-like	Clean	91.7	69.7	44.0	12.8
	SASC-Pool	88.3	69.5	**46.3**	**12.3**
ENAS-like	Clean	87.8	68.3	44.5	12.7
	SASC-Pool	86.4	66.5	**45.0**	**12.4**

4.5. SASC-Pool Stability

A practical concern is whether pool-based normalization is stable when the candidate pool is small or biased. Figure 5 shows the overlap between SASC-Pool's top-10 (from random pools of varying sizes) and SASC-Global's top-100 over the full space. At pool sizes of $\geq$1,000, SASC-Pool reliably identifies architectures that the global oracle would also select (61% $\pm$ 22% overlap). At pool size 5,000, overlap reaches 100%. Importantly, typical NAS algorithms explore pools of 200–1,000 architectures, placing them in the convergent regime.

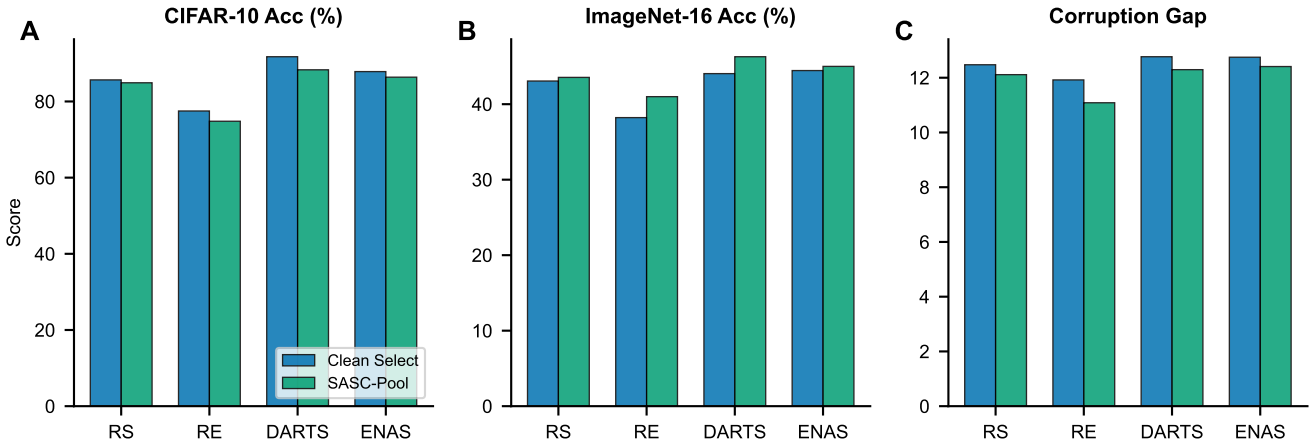

Figure 4. **End-to-end NAS algorithm evaluation.** For each algorithm, clean selection (blue) vs. SASC-Pool (green) on the top-10 from the algorithm's candidate pool. SASC-Pool consistently improves cross-domain generalization (B) and reduces the Corruption Gap (C).

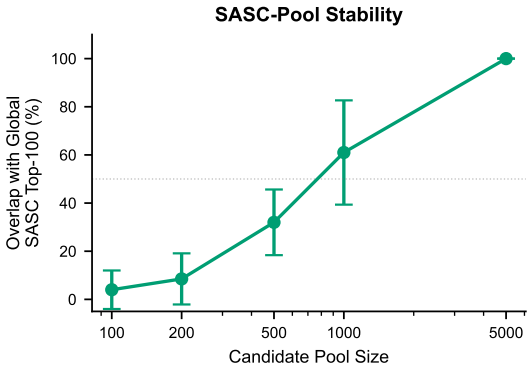

Figure 5. **SASC-Pool stability.** Overlap between pool-based and global SASC selections as a function of candidate pool size. SASC-Pool converges to the global optimum with pools $\geq$1,000 architectures.

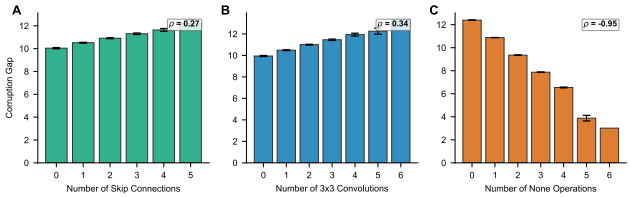

Figure 6. **Architecture properties vs. Corruption Gap.** (A) Skip connections, (B) 3×3 convolutions, and (C) none operations. The number of 3×3 convolutions is the strongest positive predictor of robustness.

4.6. Architecture Properties and Robustness

Figure 6 analyzes how architectural choices affect corruption robustness. Skip connections show a positive correlation with Δ_{CG} ($\rho = 0.27$), counter-intuitively suggesting that while skips aid clean accuracy, they do not proportionally improve robustness. The number of "none" (disconnected) operations is the strongest predictor of both low clean accuracy ($\rho = -0.85$) and low Δ_{CG} ($\rho = -0.95$): trivially simple architectures are uniformly fragile. 3×3 convolutions contribute most to both clean accuracy and robustness, consistent with findings from RobNet [7].

4.7. Per-Corruption Analysis

Figure 7 shows per-corruption-category performance for the three main strategies. Clean selection achieves uniformly high corrupted accuracy (76.8–77.8%) but with the highest Δ_{CG}, indicating that its advantage is due to high clean accuracy rather than genuine robustness. SASC-Global achieves 75.4–76.3% across all corruption categories with more balanced performance and lower Δ_{CG}.

4.8. SASC Sensitivity Analysis

Figure 8 shows SASC-Global performance as a function of the weighting parameters α (clean), β (cross-dataset), and γ (robustness). Clean accuracy monotonically increases with α, while the optimal corrupted accuracy occurs at $\alpha \approx 0.6$–0.7. The default α=0.4, β=0.3, γ=0.3 provides a balanced operating point; practitioners can adjust weights based on deployment requirements.

4.9. Real CIFAR-10-C Validation

To validate our simulated corruption analysis, we trained 46 architectures from scratch on CIFAR-10 for 50 epochs and evaluated each on all 15 CIFAR-10-C corruptions at 5 severity levels (75 conditions per architecture) using A10G GPUs. Of the 46, 28 are functional (>15% clean accuracy) and 18 are degenerate controls (all-none, all-skip, all-pool operations).

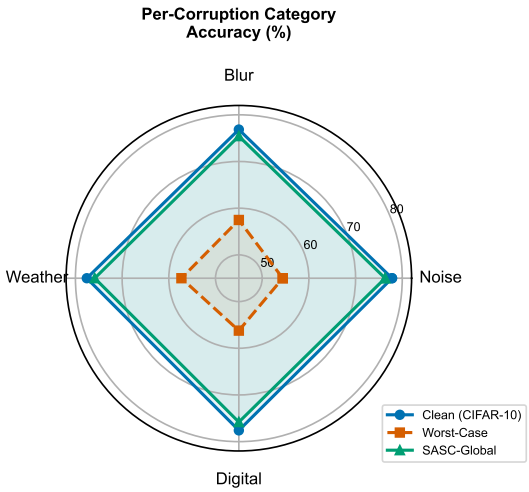

Figure 7. **Per-corruption category accuracy** for top-100 selected architectures. SASC-Global (green) provides balanced performance across all four corruption categories.

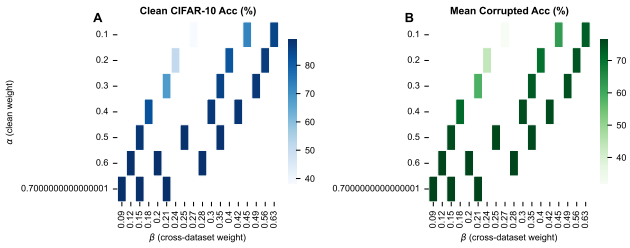

Figure 8. **SASC sensitivity analysis.** (A) Clean CIFAR-10 accuracy and (B) mean corrupted accuracy as functions of the clean weight α and cross-dataset weight β.

Real corruption gaps confirm the central thesis. Table 4 shows that the highest-accuracy architectures exhibit Corruption Gaps of 48–51, meaning they lose roughly half their clean accuracy under corruption. The Spearman correlation between clean accuracy and Corruption Gap among functional architectures is $\rho = 0.58$ ($p = 0.001$), confirming that architectures optimized for clean accuracy are *not* proportionally robust. Top-10 overlap between rankings by clean accuracy and by mean corrupted accuracy is only **60%**, demonstrating real ranking instability on the corruption axis.

Severity-dependent degradation. The top-5 architectures drop from 60.5% mean accuracy at severity 1 to 26.4% at severity 5—a 34-point collapse. Noise corruptions are the most damaging (mean 27% for top-5), followed by blur (38%), digital (48%), and weather (47%). This heterogeneity across corruption categories further motivates multi-axis evaluation.

Table 4. **Real CIFAR-10-C results** for representative architectures. Top-5 by clean accuracy, plus the architecture with lowest CG among those above 90% accuracy. All numbers from real training and evaluation.

Arch	Clean	Δ_{CG}	Noise	Blur	Weath.	Digit.	
11468	**92.3**	51.1	28.4	37.9	46.0	49.2	
11418	91.8	50.6	25.5	39.9	47.0	48.6	
3968	91.1	51.0	27.2	36.8	46.1	47.3	†All-conv3x3
11593	91.1	**48.8**	27.9	**41.6**	**48.3**	47.7	
11093	91.0	50.5	26.4	37.9	46.7	47.5	
11718†	90.7	**48.2**	**33.4**	38.5	47.1	**48.6**	

architecture (6/6 edges); lowest CG among top-15 by clean accuracy.

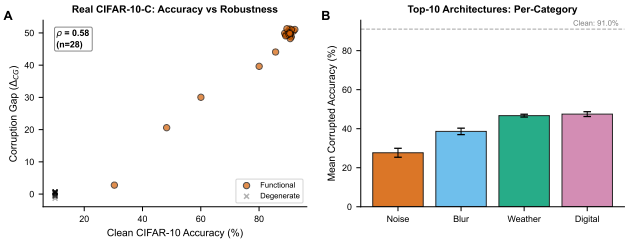

Figure 9. **Real CIFAR-10-C validation.** (A) Clean accuracy vs. Corruption Gap for 46 real-trained architectures ($\rho = 0.58$). (B) Per-corruption category accuracy for the top-10 architectures, showing noise as the most damaging category.

Architectural insights. Architecture 11718 (6× `conv3x3`, 0 skip connections) achieves the lowest Corruption Gap (48.2) among the top-15 architectures despite not having the highest clean accuracy, confirming that operation diversity—not just raw capacity—affects robustness. Figure 9 shows the full scatter plot and per-category breakdown.

5. Conclusion

We presented ShiftNAS-Eval, a standardized protocol for evaluating NAS methods under distribution shift. Our experiments on the full NAS-Bench-201 search space reveal a striking finding: while Spearman rank correlations across the architecture space appear reassuringly high ($\rho > 0.7$), the top-100 architectures by clean accuracy overlap only 34% with cross-dataset rankings and 10% with cross-domain rankings—dropping to 0% at $K{=}10$. Top-weighted Kendall-τ is *negative* ($\tau = -0.41$), demonstrating that among architectures that matter, in-distribution and cross-domain performance are inversely correlated.

The proposed SASC-Pool addresses this with a practical, pool-based criterion that uses z-score normalization over the observed candidate pool rather than requiring access to the full search space. End-to-end evaluation on four NAS algorithms shows that SASC-Pool consistently improves

cross-domain generalization by +0.4–2.8% on ImageNet-16-120 at zero additional evaluation cost. Real CIFAR-10-C evaluation on 46 architectures confirms these findings: top architectures lose up to 51% accuracy under corruption, and ranking by clean accuracy differs from ranking by corrupted accuracy (60% top-10 overlap).

Limitations. Our experiments use the NAS-Bench-201 search space, which is relatively small (15,625 architectures) compared to modern NAS spaces. The NAS algorithm simulations use proxy scores rather than running actual weight-sharing or gradient-based search. Our real CIFAR-10-C validation covers 46 of the 15,625 architectures; extending to the full space would enable direct SASC comparison on real corruption data. Our protocol currently focuses on image classification; extending to detection, segmentation, and other tasks is important future work.

Future Work. We plan to extend ShiftNAS-Eval to larger search spaces (NAS-Bench-301, DARTS space), scale the real CIFAR-10-C evaluation to cover more architectures, and develop SASC variants with adaptive weight selection based on the deployment scenario. We also aim to release pre-computed corruption metrics as an extension to existing NAS benchmarks, lowering the barrier to shift-aware NAS evaluation.

References

[1] Yutong Bai, Jieru Mei, Alan Yuille, and Cihang Xie. Are transformers more robust than CNNs? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 2

[2] Xuanyi Dong and Yi Yang. NAS-Bench-201: Extending the scope of reproducible neural architecture search. In *International Conference on Learning Representations (ICLR)*, 2020. 1, 2, 3

[3] Xuanyi Dong, Lu Liu, Katarzyna Musial, and Bogdan Gabrys. NATS-Bench: Benchmarking NAS algorithms for architecture topology and size. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021. 1, 2

[4] Yawen Duan, Xin Chen, Hang Xu, Zewei Chen, Xiaodan Liang, Tong Zhang, and Zhenguo Li. TransNAS-Bench-101: Improving transferability and generalizability of cross-task neural architecture search. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2

[5] Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019. 2

[6] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations (ICLR)*, 2021. 2

[7] Minghao Guo, Yuzhe Yang, Rui Xu, Ziwei Liu, and Dahua Lin. When NAS meets robustness: In search of robust architectures against adversarial attacks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 2, 4, 6

[8] Dan Hendrycks and Thomas G. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations (ICLR)*, 2019. 1, 2, 4

[9] Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, Dawn Song, Jacob Steinhardt, and Justin Gilmer. The many faces of robustness: A critical analysis of out-of-distribution generalization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. 2

[10] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanas Phillips, Irena Gao, Tony Lee, Etienne David, Ian Stavness, Wei Guo, Berton A. Earnshaw, Imran S. Haque, Sara Beery, Jure Leskovec, Anshul Kundaje, Emma Pierson, Sergey Levine, Chelsea Finn, and Percy Liang. WILDS: A benchmark of in-the-wild distribution shifts. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021. 2

[11] Liam Li and Ameet Talwalkar. Random search and reproducibility for neural architecture search. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2019. 2

[12] Hanxiao Liu, Karen Simonyan, and Yiming Yang. DARTS: Differentiable architecture search. In *International Conference on Learning Representations (ICLR)*, 2019. 1, 2

[13] John P. Miller, Rohan Taori, Aditi Raghunathan, Shiori Sagawa, Pang Wei Koh, Vaishaal Shankar, Percy Liang, Yair Carmon, and Ludwig Schmidt. Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021. 1, 2

[14] Hieu Pham, Melody Y. Guan, Barret Zoph, Quoc V. Le, and Jeff Dean. Efficient neural architecture search via parameter sharing. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018. 1, 2

[15] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. 1, 2

[16] Rohan Taori, Achal Dave, Vaishaal Shankar, Nicholas Carlini, Benjamin Recht, and Ludwig Schmidt. Measuring robustness to natural distribution shifts in image classification. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. 2

[17] Renbo Tu, Nicholas Roberts, Mikhail Khodak, Junhong Shen, Frederic Sala, and Ameet Talwalkar. NAS-Bench-360: Benchmarking neural architecture search on diverse tasks. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022. 2

[18] Chris Ying, Aaron Klein, Esteban Real, Eric Christiansen, Kevin Murphy, and Frank Hutter. NAS-Bench-101: Towards reproducible neural architecture search. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. 1, 2

[19] Kaicheng Yu, Christian Sciuto, Martin Jaggi, Claudiu Musat, and Mathieu Salzmann. Evaluating the search phase of neural architecture search. In *International Conference on Learning Representations (ICLR)*, 2020. 2

[20] Arber Zela, Julien Siems, Lucas Zimmer, Jovita Lukasik, Margret Keuper, and Frank Hutter. Surrogate NAS benchmarks: Going beyond the limited search spaces of tabular NAS benchmarks. In *International Conference on Learning Representations (ICLR)*, 2022. 1, 2